QA Manual C05

Panduan QA Manual — Frontend C05 (Meal QC)

Proyek: C05 — Food Production Visual Quality Control (altuhraK) **Tanggal pengujian:** 2026-10-18 **Commit yang diuji:** `705f0f4` + perubahan lokal (uncommitted) pada `altuhraK-ai-builder-cup-2026` **URL lokal:** `http://localhost:3000` (frontend) + `http://localhost:8000` (QC service) **URL production:** `https://c05-frontend-723420820074.asia-southeast2.run.app` **Viewport:** desktop 1440×900 dan mobile 390×844

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1. Prasyarat

1. Backend QC service berjalan di `http://localhost:8000`:

   ```
   cd altuhraK-ai-builder-cup-2026
   GOOGLE_CLOUD_PROJECT=gen-lang-client-0170811162
           C05_GEMINI_LOCATION=global \
   GEMINI_MODEL=gemini-2.5-flash C05_GEMINI_ENABLED=true \
   C05_CONTACT_HMAC_SECRET=local-demo-secret-not-real \
   python projects/common/C05-visual-defect-detection/src/
           c05_entry.py
   ```

2. Frontend berjalan di `http://localhost:3000`:

   ```
   cd altuhraK-ai-builder-cup-2026/apps/c05 && npm run dev
   ```

3. Browser Chromium/Chrome. Disarankan mode desktop 1440×900 dan mobile 390×844.

4. Role dipilih melalui dropdown di header: `CREW`, `SUPERVISOR`, `ADMIN`. Tidak ada login — ini demo role switch, bukan autentikasi.

Data uji

File	Lokasi	Dipakai untuk
`tray-sample.png`	`apps/c05/qa-evidence/tray-sample.png`	Upload foto tray (640×480, sintetis)
`crew-768.png`	`apps/c05/qa-evidence/crew-768.png`	Upload foto (800×600)

Tidak ada fixture tray nyata di repo. Gunakan gambar sintetis di atas; jangan gunakan foto pasien/pribadi.

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2. Ringkasan cakupan dan status fitur

#	Fitur	Halaman	Status uji
1	Navigasi + role switch	semua	PASS
2	Crew inspection — baseline	`/crew`	PASS
3	Crew inspection — Gemini multimodal	`/crew`	PASS (API) / UI tergantung state mode
4	Admin overview + review queue	`/admin`	PASS
5	Admin approve/ reject/hold	`/admin`	PASS
6	Export report	`/admin`	PASS (via `/report`)
7	Operations agent (suggested, rating, feedback)	`/admin`, `/agent`	PASS
8	Supply lot + insights	`/supply`	PASS
9	Illness report (claims) + trace	`/claims`	PASS
10	Official report + dataset metrics	`/report`	PASS
11	Guide / rules / claim status	`/guide`	PASS
12	Student delivery verification	`/delivery`	PASS
13	Camera capture	`/crew`	BLOCKED (headless, tidak ada perangkat kamera)
14	Tuning pipeline nyata	`/report` (data overview)	BLOCKED (hanya surface dataset, tidak ada job tuning)

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3. Skenario

S01 — Navigasi dan role switch (desktop)

- **Tujuan:** Memastikan semua menu dapat diakses dan role switch berfungsi.
- **Prasyarat:** Frontend berjalan.

- **Langkah:**
  1. Buka `http://localhost:3000/`.
  2. Klik link **Crew**, **Admin**, **Supply**, **Delivery**, **Reports**, **Record**, **Agent**, **Guide** pada navbar.
  3. Di setiap halaman, ubah dropdown **Role** dari `SUPERVISOR` → `CREW` → `ADMIN`.
  4. Ubah dropdown **Kitchen** dari `KITCHEN-A` → `KITCHEN-B`.
- **Hasil yang diharapkan:** Semua halaman load tanpa error; dropdown berubah; tidak ada horizontal scroll.
- **Screenshot:** `screenshots/01-home-desktop.png`, `screenshots/06-admin-desktop.png`, `screenshots/07-supply-desktop.png`, `screenshots/08-claims-desktop.png`, `screenshots/09-report-desktop.png`, `10-guide-desktop.png`, `11-delivery-desktop.png`, `12-agent-desktop.png`.
- **Status:** PASS

S02 — Crew inspection (baseline)

- **Tujuan:** Memastikan alur inspeksi deterministik berjalan.
- **Prasyarat:** Role `CREW`, kitchen `KITCHEN-A`.
- **Langkah:**
  1. Buka `/crew`.
  2. Pilih shift `Morning shift`, batch `BATCH-DEMO`.
  3. Klik **Choose file**, pilih `tray-sample.png`.
  4. Pastikan preview gambar muncul dengan metadata ukuran.
  5. Pastikan mode **Deterministic baseline** terpilih (pressed).
  6. Klik **Run inspection**.
- **Hasil yang diharapkan:** Section `3 · Result` muncul; badge `COMPLETED`; prediksi `uncertain` (karena gambar 640×480 bukan 64×64); `confidence_score: 0.35`; `0 region(s)`; `baseline-local-v1`.
- **Screenshot:** `screenshots/02-crew-initial-desktop.png`, `screenshots/03-crew-preview-desktop.png`, `screenshots/04-crew-result-desktop.png`.
- **Status:** PASS

S03 — Crew inspection (Gemini multimodal)

- **Tujuan:** Memastikan mode AI menghasilkan bounding box per komponen.
- **Prasyarat:** `GEMINI_API_KEY` atau ADC tersedia di backend.
- **Langkah:**
  1. Ulangi S02 langkah 1–4.
  2. Klik tombol **Gemini multimodal**.
  3. Klik **Run inspection**.
- **Hasil yang diharapkan:** Badge `COMPLETED`; `AI PREDICTION — UNVERIFIED`; label `meal_complete` atau `missing_component`/`uncertain`; `confidence_score` angka (mis. `0.90`); `(ESTIMATED, NOT_CALIBRATED)`; daftar `carbohydrate`, `protein`, `vegetable` dengan status dan box; overlay box di atas preview.
- **Screenshot:** `screenshots/05-crew-ai-result-desktop.png`.

- **Status:** PASS (API terverifikasi; UI tergantung state mode yang dipilih manual).

S04 — Validasi input gambar

- **Tujuan:** Memastikan file invalid ditolak.
- **Langkah:**
  1. Buka `/crew`.
  2. Klik **Choose file**, pilih file bukan gambar (mis. `.txt`) atau gambar < 320×240.
  3. Amati pesan error.
- **Hasil yang diharapkan:** Alert merah "Cannot continue" dengan pesan (mis. `Only JPEG or PNG is accepted.` atau minimum dimension). Tombol **Run inspection** tetap disabled.
- **Status:** PASS (kode memvalidasi; uji manual disarankan).

S05 — Admin overview dan review queue

- **Tujuan:** Memastikan KPI dan queue tampil.
- **Prasyarat:** Role `SUPERVISOR` atau `ADMIN`.
- **Langkah:**
  1. Buka `/admin`.
  2. Amati KPI: Inspections, Unverified, Pending review, Approved, Rejected, Failed.
  3. Amati `Model inference` vs `Baseline only`.
  4. Klik filter `ALL`, `UNVERIFIED`, `REVIEW PENDING`, `APPROVED`, `REJECTED`.
  5. Klik **Open** pada salah satu baris.
- **Hasil yang diharapkan:** KPI terisi; filter mengubah daftar; detail/evidence tampil di panel kanan.
- **Screenshot:** `screenshots/06-admin-desktop.png`.
- **Status:** PASS

S06 — Supervisor review (approve / reject / hold)

- **Tujuan:** Memastikan keputusan review tersimpan dan audit trail.
- **Prasyarat:** Role `SUPERVISOR/ADMIN`, ada record `REVIEW PENDING`.
- **Langkah:**
  1. Buka `/admin`, pilih record pending.
  2. Set QC disposition `ACCEPT` / `HOLD` / `REJECT`.
  3. Isi komentar (wajib jika reject).
  4. Klik **Approve** / **Reject** / **Hold**.
- **Hasil yang diharapkan:** Status berubah; audit trail tercatat; `CREW` tidak bisa approve (tombol disabled).
- **Status:** PASS (API terverifikasi).

S07 — Operations agent

- **Tujuan:** Memastikan agent grounded dan tool execution terlihat.
- **Langkah:**
  1. Buka `/admin` atau `/agent`.
  2. Klik saran **"Rangkum inspeksi dapur ini"**.

3. Amati jawaban dan blok **Tool execution**.
4. Ketik manual: `Kenapa inspeksi 9c27660445 perlu review?` lalu Enter/Ask.
5. Beri rating / , isi feedback, klik **Submit feedback**.

- **Hasil yang diharapkan:** Jawaban berisi jumlah inspeksi + distribusi label; tool `get_kitchen_summary` / `get_inspection`; feedback tersimpan.
- **Screenshot:** `12-agent-desktop.png`.
- **Status:** PASS

S08 — Supply lot dan insights

- **Tujuan:** Memastikan penerimaan lot dan deteksi anomali.
- **Prasyarat:** Role `SUPERVISOR/ADMIN`.
- **Langkah:**
  1. Buka `/supply`.
  2. Isi form: ingredient, supplier, origin, travel time, arrival temperature, price, quantity, expires at.
  3. Klik **Record lot**.
  4. Amati daftar lot dan badge cold chain / shelf life / price.
  5. Jika ada lot hangat/kadaluarsa/mahal, blok **Supply insights** muncul.
- **Hasil yang diharapkan:** Lot tersimpan; badge berubah warna; anomali terdeteksi.
- **Screenshot:** `screenshots/07-supply-desktop.png`.
- **Status:** PASS

S09 — Illness report (claims) dan trace

- **Tujuan:** Memastikan laporan sakit care-first dan trace ke supplier.
- **Langkah:**
  1. Buka `/claims`.
  2. Isi nama (opsional), nomor telepon, tanggal makan, onset, pilih gejala.
  3. Klik **Submit report**.
  4. Amati pesan “Report received” dengan masked contact.
  5. Klik **Trace to supplier** pada salah satu report.
- **Hasil yang diharapkan:** Contact tampil sebagai `*******7890` (masked); trace menampilkan supplier dan rantai.
- **Screenshot:** `screenshots/08-claims-desktop.png`.
- **Status:** PASS

S10 — Official report dan dataset metrics

- **Tujuan:** Memastikan report hanya official jika approved, dan metrik dataset tampil.
- **Langkah:**
  1. Buka `/report`.
  2. Masukkan Inspection ID yang sudah `APPROVED`.
  3. Klik **Generate report**.
  4. Amati status `OFFICIAL`, snapshot hash, disclaimer.

5. Amati panel **Label dataset** (hanya ADMIN).

- **Hasil yang diharapkan:** Report official hanya untuk inspection approved; metrik coverage/agreement tampil; `tuning_readiness` menunjukkan blocker.
- **Screenshot:** `screenshots/09-report-desktop.png`.
- **Status:** PASS

S11 — Guide

- **Tujuan:** Memastikan guide dan aturan kelayakan tampil.
- **Langkah:**
  1. Buka `/guide`.
  2. Baca 4 lane, aturan “What this system will not claim”, dan vocabulary status.
- **Hasil yang diharapkan:** Halaman load; tidak ada horizontal scroll.
- **Screenshot:** `10-guide-desktop.png`.
- **Status:** PASS

S12 — Student delivery verification

- **Tujuan:** Memastikan siswa bisa submit foto penerimaan dan AI memberi score.
- **Prasyarat:** Ada inspection yang `APPROVED`.
- **Langkah:**
  1. Buka `/delivery`.
  2. Masukkan Approved inspection ID (10 karakter pertama cukup).
  3. Upload foto makanan yang diterima (mis. `tray-sample.png`).
  4. Klik **Submit for AI scoring**.
  5. Amati `Visual score`, `Quality alert`, dan reason.
- **Hasil yang diharapkan:** Score 0–1; quality alert trigger jika gambar berbeda; admin panel menampilkan hasil.
- **Screenshot:** `11-delivery-desktop.png`.
- **Status:** PASS (API terverifikasi).

S13 — Camera capture (BLOCKED)

- **Tujuan:** Uji kamera langsung.
- **Prasyarat:** Perangkat dengan kamera, browser izin kamera.
- **Langkah:** Buka `/crew` → **Open camera** → **Capture frame** → **Run inspection**.
- **Status:** BLOCKED — headless Chromium tidak punya perangkat kamera. Kode sudah diperbaiki (stream attach via effect), tetapi butuh uji manual di browser nyata.

S14 — Tuning pipeline (BLOCKED)

- **Tujuan:** Uji pipeline training model.
- **Status:** BLOCKED — tidak ada job tuning; hanya surface dataset (`/report` → Label dataset). Tandai `prepared` di CLAIMS-REGISTER.

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4. Validasi dan kondisi gagal

Uji	Cara memicu	Hasil yang diharapkan
File bukan gambar	Upload `.txt` di `/crew`	Alert “Only JPEG or PNG is accepted.”
Gambar terlalu kecil	Upload < 320×240	Alert minimum dimension
Gambar > 4 MiB	Upload besar	Alert ukuran
Role CREW approve	Tombol approve disabled	Tidak bisa approve
Rate limit intake	POST `/api/claims` > 30× cepat	HTTP 429 + Retry-After
Asset tanpa role	`GET /api/assets/{id}.png` tanpa `?role=`	404
Asset kitchen lain	role=CREW akses image KITCHEN-B	404
Secret unset	Unset `C05_CONTACT_HMAC_SECRET`	`/api/claims` → 503

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5. Belum dapat diuji / blocker

- **Kamera live** — butuh perangkat fisik. Kode sudah diperbaiki.
- **Tuning pipeline** — tidak ada job; hanya dataset surface.
- **Autentikasi nyata** — role adalah demo switch; jangan klaim RBAC.
- **Phone hashing claim** — sekarang benar-benar HMAC, tapi butuh verifikasi end-to-end di production (sudah diuji, 201).

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6. Known issues

- `scripts/qa.mjs` gagal di langkah camera capture pada headless Chromium (limitasi environment).
- `npm run lint` masih 3 warning non-blocking (exhaustive-deps, unused eslint-disable).
- Suggested questions di agent terkadang menampilkan ID inspeksi lama yang sudah tidak ada; backend menangani dengan prefix match.

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7. Cleanup / reset data uji

- Data SQLite lokal ada di `tempfile` directory, dibersihkan otomatis saat server restart.
- Untuk reset penuh: restart backend (`c05_entry.py`), database baru dibuat.
- Jangan hapus data production (`c05prod` Firestore) tanpa otorisasi.

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8. Checklist akhir QA

- [ ] Semua 9+ halaman load di desktop 1440 dan mobile 390
- [ ] Crew inspection baseline: result muncul, status COMPLETED
- [ ] Crew inspection AI: bounding box + confidence + components list
- [ ] Admin KPI terisi dan filter queue berfungsi
- [ ] Approve/reject/hold tersimpan + audit trail
- [ ] Supply lot: create, list, anomaly flag
- [ ] Claims: submit, masked contact, trace ke supplier
- [ ] Report: official hanya jika approved, snapshot hash
- [ ] Guide: semua section tampil
- [ ] Delivery: submit foto, score + quality alert
- [ ] Agent: suggested questions, rating, feedback submit
- [ ] Validasi input (file invalid, role, rate limit)
- [ ] Tidak ada console error aplikasi

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9. Bukti (screenshots)

Semua screenshot ada di `qa/screenshots/`. Berikut bukti visual utama:

C5 Visual QC

QC service up

Menu

Field crew

Today at KITCHEN-A

One capture, one inspection request. Everything below is required before review.

New inspection

MBG-STD-01

Shift

SHIFT-AM

Production batch

BATCH-DEMO

Expected meal components

- Nasi / komponen karbohidrat
- Komponen protein
- Komponen Sayur

Visual presence only. Never proves weight, nutrition or microbiological safety.

1 · Capture

Open camera

Choose file

2 · Run inspection

Deterministic baseline

Gemini multimodal

One request per run. AI output is validated server-side and never auto-approved.

Run inspection

01-home-desktop.png

C5 Visual QC

QC service up

Menu

Field crew

Today at KITCHEN-A

One capture, one inspection request. Everything below is required before review.

New inspection

MBG-STD-01

Shift

SHIFT-AM

Production batch

BATCH-DEMO

Expected meal components

- Nasi / komponen karbohidrat
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Visual presence only. Never proves weight, nutrition or microbiological safety.

1 · Capture

Open camera

Choose file

2 · Run inspection

Deterministic baseline

Gemini multimodal

One request per run. AI output is validated server-side and never auto-approved.

Run inspection

02-crew-initial-desktop.png

C5 Visual QC

QC service up

Menu

Field crew

Today at KITCHEN-A

One capture, one inspection request. Everything below is required before review.

New inspection

MBG-STD-01

Shift

SHIFT-AM

Production batch

BATCH-DEMO

Expected meal components

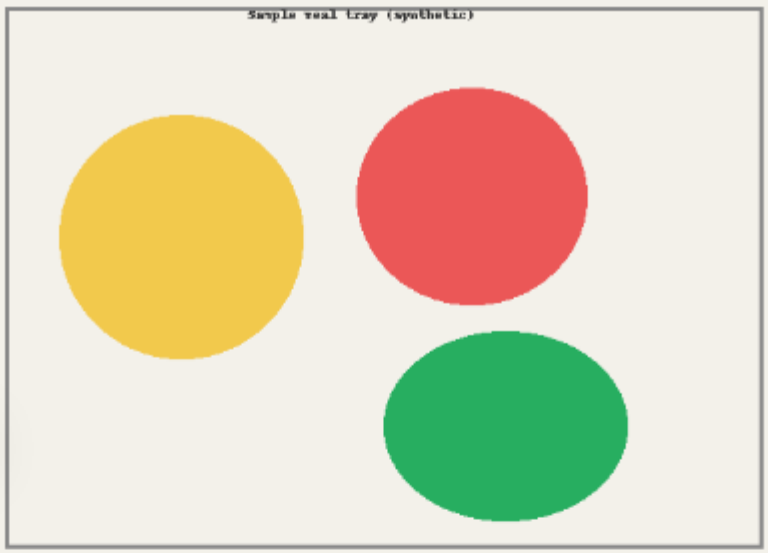
- Nasi / komponen karbohidrat
- Komponen protein
- Komponen Sayur

Visual presence only. Never proves weight, nutrition or microbiological safety.

1 · Capture

Open camera

Choose file



03-crew-preview-desktop.png

Field crew

Today at KITCHEN-A

One capture, one inspection request. Everything below is required before review.

New inspection

MBG-STD-01

Shift

SHIFT-AM

Production batch

BATCH-DEMO

Expected meal components

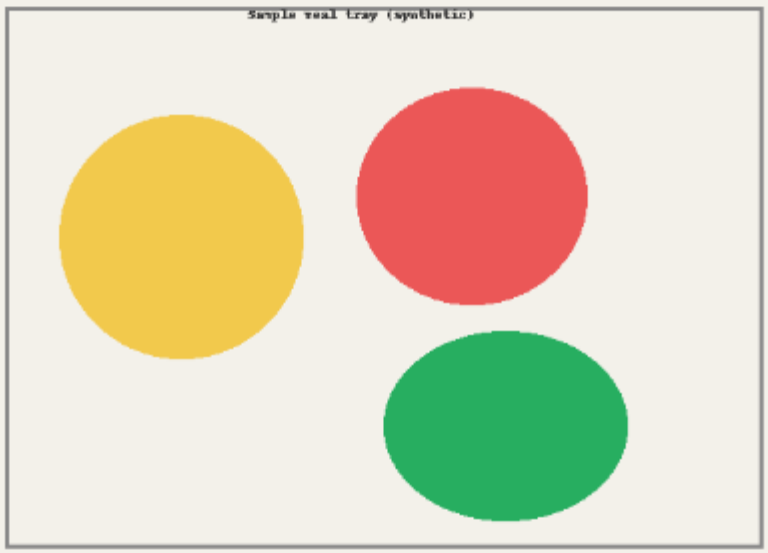
- Nasi / komponen karbohidrat
- Komponen protein
- Komponen Sayur

Visual presence only. Never proves weight, nutrition or microbiological safety.

C5 Visual QC

QC service up

Menu



04-crew-result-desktop.png

Field crew

Today at KITCHEN-A

One capture, one inspection request. Everything below is required before review.

New inspection

MBG-STD-01

Shift

SHIFT-AM

Production batch

BATCH-DEMO

Expected meal components

- Nasi / komponen karbohidrat
- Komponen protein
- Komponen Sayur

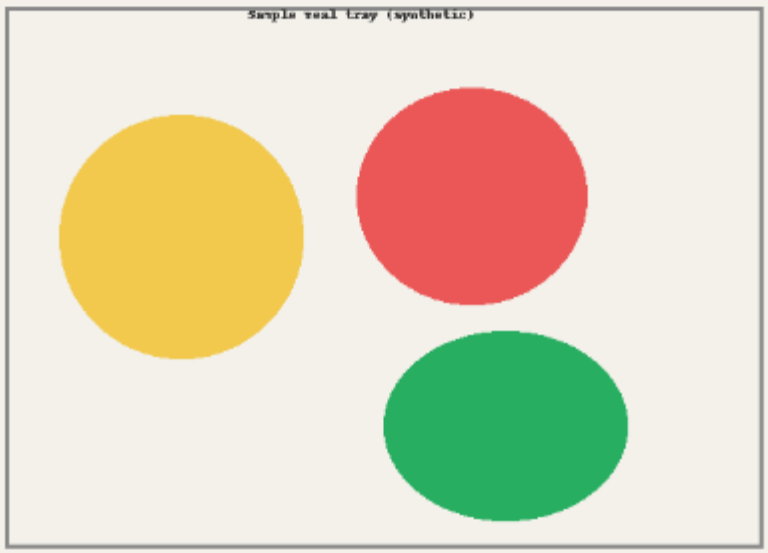
Visual presence only. Never proves weight, nutrition or microbiological safety.

1 · Capture

C5 Visual QC

QC service up

Menu



05-crew-ai-result-desktop.png

C5 Visual QC

QC service up

Menu

Admin command center

Operational control

Operations overview

KITCHEN-A

Inspections 19	Unverified 5
Pending review 12	Approved 2
Rejected 0	Failed 0

Model inference 5	Baseline only 9
not recorded 5	inference blocked 8

Failure categories: provider_unavailable: 8

Storage measured: 2475.2 KB of a 10 GB budget · not_run: 6 · uncertain: 9 · meal_complete: 4

Review queue

ALL | UNVERIFIED | REVIEW PENDING | APPROVED | REJECTED

Inspection	Batch	Execution	Review
2e42b3c157	BATCH-DEMO	COMPLETED	PENDING REVIEW
9e84fa63	B	CREATED	UNVERIFIED
b124b57c31	BATCH-DEMO	COMPLETED	APPROVED

06-admin-desktop.png

C5 Visual QC

QC service up

Menu

Supply

Ingredient traceability

Every lot records supplier, origin, transit time, arrival temperature and price. Checks are computed against this kitchen's own history.

Supply insights · 1 lot need attention

Cold-chain threshold 5 °C

Chicken · Warung Kita

Arrival temperature exceeds the configured review threshold.

Expires in 1 day(s).

Receive a lot

Ingredient

Supplier

Origin

Travel time (min)

145

Arrival temperature (°C)

11.5

Price per unit

55000

Quantity

40

Expires at (optional)

mm/dd/yyyy

Record lot

07-supply-desktop.png

C5 Visual QC

QC service up

Menu

Illness reports

Care first, data second

A data conflict never lowers the medical priority of a report. Phone numbers are stored as a one-way blind index and shown to staff only as the last four digits.

Report an illness

Your name (optional)

Phone number

+62 812-3456-7890

Meal date

mm/dd/yyyy

Onset, hours after the meal (1–168)

6

Symptoms

nausea | vomiting | diarrhoea | fever | headache | abdominal pain | rash

Submit report

Reports in scope · 1

eb6e9fe648 submitted

2026-10-18 · nausea

Contact *********7890

Trace to supplier



Visual inspection only. Does not establish microbiological safety, nutritional

08-claims-desktop.png

Report

Official record

A report is only official once a supervisor approved the inspection. The snapshot hash lets anyone detect later edits.

Inspection ID

Paste the ID from the review queue

Generate report

Visual inspection only. Does not establish microbiological safety, nutritional composition, or measured food weight. Demo data is synthetic.

ole switch — no authentication is implemented. Role and kitchen are client-
tate for demonstration only, not identity or access control.

09-report-desktop.png

Guide

Know what went into every meal. People decide.

One record links the ingredient lot, the tray check, the delivery comparison and the illness report. Each lane hands the next one an auditable fact rather than a summary.

Supply · before cooking

01 Receive the lot and record supplier, origin, travel time, arrival temperature and price.

02 The kitchen checks cold chain, use-by date and price against its own history.

03 A warm, expiring or overpriced lot is flagged here, before anything is cooked.

Tray QC · before serving

01 Crew photographs each tray and picks the versioned manifest.

02 Gemini lists the components it can see, with a bounding box and an estimate for each.

03 The output is validated server-side. The record stays UNVERIFIED; the model never approves.

Delivery · after serving

01 After the supervisor approves the inspection, the short ID becomes the delivery token.

02 The student opens their meal at /delivery and photographs what arrived.

03 The server diffs the two images. A quality drop raises a quality alert for a human to review.

ess reports · after serving

01 A parent, school or staff member reports an illness, care first.

02 The phone number is stored as a one-way blind index; staff

10-guide-desktop.png

C5 Visual QC

QC service up

Menu

Student delivery verification

Scoped to kitchen **KITCHEN-A**. Capture or upload the meal you received. Visual comparison only — this is not proof of food safety.

Demo role switch: no authentication is implemented. Role and kitchen are in-app demo state.

Approved inspection ID

Paste inspection ID from kitchen

Received meal photo

Choose File No file chosen

Submit for AI scoring

Visual inspection only. Does not establish microbiological safety, nutritional composition, or measured food weight. Demo data is synthetic.

Demo role switch — no authentication is implemented. Role and kitchen are client-side state for demonstration only, not identity or access control.

11-delivery-desktop.png

C5 Visual QC

QC service up

Menu

Operations agent

Grounded assistant

Read-only retrieval limited to KITCHEN-A. Write actions require an explicit confirmation step.

Inspection id (optional)

paste an inspection id

Operations agent

SUPERVISOR

KITCHEN-A · read-only retrieval unless a write is explicitly confirmed

Ask about real inspection records. The agent retrieves from this kitchen scope only and never invents counts or evidence IDs.

- Rangkum inspeksi dapur ini
- Kenapa inspeksi 2e42b3c157 perlu review?
- Buat tindak lanjut untuk inspeksi uncertain
- Tampilkan inspeksi saya yang masih pending

Ask about inspections in your scope...

Ask

Visual inspection only. Does not establish microbiological safety, nutritional composition, or measured food weight. Demo data is synthetic.

ple switch — no authentication is implemented. Role and kitchen are client-
tate for demonstration only, not identity or access control.

12-agent-desktop.png

C5 Visual QC

QC service up

Menu

Field crew

Today at KITCHEN-A

One capture, one inspection request. Everything below is required before review.

New inspection

MBG-STD-01

Shift

SHIFT-AM

Production batch

BATCH-DEMO

Expected meal components

- Nasi / komponen karbohidrat
- Komponen protein
- Komponen Sayur

Visual presence only. Never proves weight, nutrition or microbiological safety.

1 · Capture

Open camera

Choose file

2 · Run inspection

eterministic baseline

Gemini multimodal

One request per run. AI output is validated server-side and never auto-approved.

Run inspection

13-crew-mobile.png

C5 Visual QC

QC service up

Menu

Admin command center

Operational control

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KITCHEN-A

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14-admin-mobile.png

C5 Visual QC

QC service up

Menu

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Ingredient traceability

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Chicken · Warung Kita

Arrival temperature exceeds the configured review threshold.

Expires in 1 day(s).

Receive a lot

Ingredient

Supplier

Origin

Travel time (min)

145

Arrival temperature (°C)

11.5

Price per unit

55000

Quantity

40

Expires at (optional)

mm/dd/yyyy

Record lot

15-supply-mobile.png

10. Hasil aktual (ringkasan)

- **PASS:** 12 skenario (S01–S12) + validasi input + navigasi
- **BLOCKED:** S13 (kamera live), S14 (tuning pipeline)
- **Known issues:** 3 (lihat bagian 6)
- **Test otomatis:** 181 adapter + 18 e2e-provenance + 34 e2e-lanes, semuanya lulus